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$$\begin{aligned} & \int \frac{x^2 + \cos^2 x}{x^2 \sin^2 x + \sin^2 x} dx \\ &= \int \frac{x^2 + \cos^2 x}{\sin^2 x (x^2 + 1)} dx \\ &= \int \frac{x^2 + 1 - \sin^2 x}{\sin^2 x (x^2 + 1)} dx \\ &= \int \frac{x^2 + 1}{\sin^2 x (x^2 + 1)} dx - \int \frac{\sin^2 x}{\sin^2 x (x^2 + 1)} dx \\ &= \int \frac{1}{\sin^2 x} dx - \int \frac{1}{x^2 + 1} dx \\ &= -\cot x - \operatorname{Bgtan} x + C \end{aligned}$$

References